

ECE253/CSE208 Introduction to Information Theory

Lecture 13: Differential Entropy

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- Chap 8 of *Elements of Information Theory (2nd Edition)* by Thomas Cover & Joy Thomas

Differential Entropy: Definition

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Definition

The differential entropy $h(X)$ of a continuous random variable X with density $f(x)$ and support $\mathcal{S}$ is defined as

$$h(X) = \mathbb{E}(-\log f(X)) = - \int_{\mathcal{S}} f(x) \log f(x) dx$$

Differential Entropy: Examples

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Example (Differential entropy can be negative)

Consider $X \sim \text{Uniform}[0, a]$, its differential entropy is

$$h(X) = - \int_0^a \frac{1}{a} \log \left(\frac{1}{a} \right) dx = \log a \implies h(X) < 0 \text{ for } 0 < a < 1$$

Example (Entropy of normal distribution)

Let $X \sim \mathcal{N}(0, \sigma^2)$ with the pdf $\phi(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{x^2}{2\sigma^2}}$. Then, we have

$$h(X) = - \int \phi(x) \ln \phi(x) dx = \frac{\mathbb{E}X^2}{2\sigma^2} + \frac{1}{2} \ln(2\pi\sigma^2) \text{ nats} = \frac{1}{2} \log(2\pi e\sigma^2) \text{ bits}$$

Note that $\log_2 x \text{ bits} = (\log_2 x) \ln 2 \text{ nats} = \ln x \text{ nats}$.

Quantization of Continuous Variable

Differential
Entropy

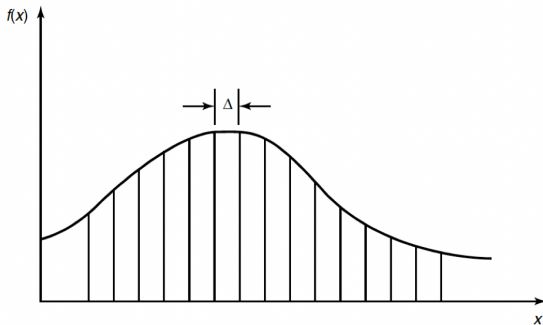
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A continuous random variable $X \sim f(x)$ and its quantized version

$$X^\Delta = x_i \text{ if } i\Delta \leq X < (i+1)\Delta,$$

where $f(x_i) \times \Delta = \int_{i\Delta}^{(i+1)\Delta} f(x) dx$.

Relate to Discrete Entropy

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Theorem

If $f(x) \log f(x)$ is Riemann integrable, then

$$H(X^\Delta) + \log \Delta \rightarrow h(X), \quad \text{as } \Delta \rightarrow 0$$

Interpretation.

- Discrete entropy of the quantized variable = differential entropy + quantization precision.
- The number of bits required to describe X with precision Δ is approximately $h(X) - \log \Delta$.

AEP for Continuous Random Variables

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Theorem

Let $X_1, X_2, \dots, X_n$ be i.i.d. with pdf $f(x)$. Then

$$-\frac{1}{n} \log f(X^n) \xrightarrow{\text{i.p.}} \mathbb{E}(-\log f(X)) = h(X).$$

Typical Set for Continuous Random Variables

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Definition (Typical set)

$$A_\epsilon^{(n)} = \{x^n \in \mathcal{S}^n : |-\frac{1}{n} \log f(x^n) - h(X)| \leq \epsilon\}, \text{ where } f(x^n) = \prod_{i=1}^n f(x_i).$$

Properties of the typical set.

1. $\Pr(A_\epsilon^{(n)}) > 1 - \epsilon$ for n sufficiently large
2. $\text{Vol}(A_\epsilon^{(n)}) \leq 2^{n(h(X)+\epsilon)}$ for all n
3. $\text{Vol}(A_\epsilon^{(n)}) \geq (1 - \epsilon)2^{n(h(X)-\epsilon)}$ for n sufficiently large,

where the volume of a set $A \subset \mathbb{R}^n$ is defined as

$$\text{Vol}(A) = \int_A dx_1 dx_2 \cdots dx_n.$$

Implications of Differential Entropy

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Theorem (cf. section 3.3 in the book)

$A_\epsilon^{(n)}$ is the smallest volume set with probability at least $1 - \epsilon$, to first order in the exponent.

Implication.

1. The volume of the smallest set that contains most of the probability $\approx 2^{nh}$.
2. The corresponding side length is $(2^{nh})^{\frac{1}{n}} = 2^h \implies h(X)$ is the logarithm of the *equivalent side length* of the smallest set that contains most of the probability.
3. A random variable with low entropy is confined to a small effective volume, and widely dispersed if it has a high entropy.
4. $h(X)$ is related to $\text{Vol}(A_\epsilon^{(n)})$. Fisher information is related to the surface area of $A_\epsilon^{(n)}$; see details in Sections 11.10 and 17.8 of Cover's book.

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Definition

$$h(X^n) = - \int f(x^n) \log f(x^n) dx^n \quad (1)$$

$$h(X|Y) = - \int f(x, y) \log f(x|y) dx dy = h(X, Y) - h(Y) \quad (2)$$

$$D_{\text{KL}}(f||g) = \int f \log \frac{f}{g} \quad (3)$$

Mutual Information

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Definition

$$I(X; Y) = \sup_{\mathcal{P}, \mathcal{Q}} I([X]_{\mathcal{P}}, [Y]_{\mathcal{Q}}) \quad (4)$$

$$= D_{\text{KL}}(f(x, y) || f(x)f(y)) \quad (5)$$

$$= h(X) - h(X|Y) \quad (6)$$

$$= h(Y) - h(Y|X) \quad (7)$$

$$= h(X) + h(Y) - h(X, Y). \quad (8)$$

The supremum is over all finite partitions $\mathcal{P}$ of X and $\mathcal{Q}$ of Y . The quantization of X by $\mathcal{P}$ is defined as $\Pr([X]_{\mathcal{P}} = i) = \Pr(X \in P_i) = \int_{P_i} f(x) dx$, where the disjoint sets P_i 's form a partition of the range of X such that $\cup_i P_i = \mathcal{X}$.

Entropy of Gaussian Distribution

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Theorem (Entropy of multivariate normal distribution)

$$h(\mathcal{N}_n(\boldsymbol{\mu}, \mathbf{K})) = \frac{1}{2} \log((2\pi e)^n |\mathbf{K}|) \text{ bits}$$

Proof: The pdf is given as

$$\phi(\mathbf{x}) = \frac{1}{(2\pi)^{n/2} |\mathbf{K}|^{1/2}} \exp\left(-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})^\top \mathbf{K}^{-1}(\mathbf{x} - \boldsymbol{\mu})\right)$$

Entropy of Gaussian Distribution: Proof

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$$\begin{aligned}h(\mathcal{N}_n(\boldsymbol{\mu}, \mathbf{K})) &= - \int \phi(\mathbf{x}) \ln \phi(\mathbf{x}) d\mathbf{x} \\&= - \int \phi(\mathbf{x}) \left[-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})^\top \mathbf{K}^{-1}(\mathbf{x} - \boldsymbol{\mu}) - \ln(2\pi)^{n/2} |\mathbf{K}|^{1/2} \right] d\mathbf{x} \\&= \frac{1}{2} \mathbb{E} [\text{Tr} (\mathbf{K}^{-1}(\mathbf{x} - \boldsymbol{\mu})(\mathbf{x} - \boldsymbol{\mu})^\top)] + \frac{1}{2} \ln(2\pi)^n |\mathbf{K}| \\&= \frac{1}{2} \text{Tr} (\mathbb{E} [\mathbf{K}^{-1}(\mathbf{x} - \boldsymbol{\mu})(\mathbf{x} - \boldsymbol{\mu})^\top]) + \frac{1}{2} \ln(2\pi)^n |\mathbf{K}| \\&= \frac{1}{2} \text{Tr} (\mathbf{K}^{-1} \mathbf{K}) + \frac{1}{2} \ln(2\pi)^n |\mathbf{K}| \\&= \frac{1}{2} \ln ((2\pi e)^n |\mathbf{K}|) \text{ nats} \\&= \frac{1}{2} \log ((2\pi e)^n |\mathbf{K}|) \text{ bits}\end{aligned}$$

Entropy of Gaussian Distribution: Example

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Example

Let $(X, Y) \sim \mathcal{N}(\mathbf{0}, \mathbf{K})$, where $\mathbf{K} = \begin{bmatrix} \sigma^2 & \rho\sigma^2 \\ \rho\sigma^2 & \sigma^2 \end{bmatrix} \implies h(X) = h(Y) = \frac{1}{2} \log(2\pi e)\sigma^2$

$$h(X, Y) = \frac{1}{2} \log((2\pi e)^2 |\mathbf{K}|) = \frac{1}{2} \log(2\pi e)^2 \sigma^4 (1 - \rho^2) \implies$$

$$I(X; Y) = h(X) + h(Y) - h(X, Y) = -\frac{1}{2} \log(1 - \rho^2) \implies$$

$$\begin{cases} X \perp Y \ (\rho = 0) & \Leftrightarrow I(X; Y) = 0 \\ X \parallel Y \ (\rho = \pm 1) & \Leftrightarrow I(X; Y) = \infty \end{cases}$$

Properties of Differential Entropy and KL Divergence

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Similar to the discrete case, we have

- $D_{\text{KL}}(f||g) \geq 0 \implies I(X; Y) \geq 0, h(X|Y) \leq h(X).$
- $h(X^n) = \sum_{i=1}^n h(X_i|X^{i-1}) \leq \sum_i h(X_i).$
- $h(X + c) = h(X)$ for any constant c .

Different from the discrete case, for a continuous random vector $\mathbf{x} \in \mathbb{R}^n$, we have

$$h(\mathbf{A}\mathbf{x}) = h(\mathbf{x}) + \log |\det(\mathbf{A})|.$$

For one dimension: $h(aX) = h(X) + \log |a|$, which can be proved by the property $f_Y(y) = \frac{1}{|a|} f_X(\frac{y}{a})$ for $Y = aX$.

Unlike discrete entropy, differential entropy is not invariant to coordinate scaling and can be negative.

Maximum Entropy

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Theorem (Normal distribution maximizes the entropy for a given covariance)

Let the random vector $\mathbf{X} \in \mathbb{R}^n$ have zero mean and covariance $\mathbf{K}$. Then,

$$\max_{\mathbb{E}(\mathbf{X}\mathbf{X}^\top)=\mathbf{K}} h(\mathbf{X}) = \frac{1}{2} \log(2\pi e)^n |\mathbf{K}|,$$

where the maximum is attained if and only if $\mathbf{X} \sim \mathcal{N}_n(\mathbf{0}, \mathbf{K})$.

Proof: Let $g(\mathbf{x})$ be any density function with zero mean and covariance $\mathbf{K}$, and $\phi(\mathbf{x}) \sim \mathcal{N}_n(\mathbf{0}, \mathbf{K})$. Then, we have

$$0 \leq D_{\text{KL}}(g||\phi) = -h(g) - \int g \log \phi = -h(g) - \int \phi \log \phi = -h(g) + h(\phi),$$

where the second equality follows since g and ϕ share the same covariance $\mathbf{K}$.

Key Step in the Maximum Entropy Proof

Note that

$$\log \phi(\mathbf{x}) = -\frac{1}{2}\mathbf{x}^\top \mathbf{K}^{-1}\mathbf{x} - \frac{1}{2}\log((2\pi)^n |\mathbf{K}|).$$

For any density $g(\mathbf{x})$ with covariance $\mathbf{K}$,

$$\mathbb{E}_g[\mathbf{x}^\top \mathbf{K}^{-1}\mathbf{x}] = \text{Tr}(\mathbf{K}^{-1}\mathbb{E}_g[\mathbf{x}\mathbf{x}^\top]) = \text{Tr}(\mathbf{K}^{-1}\mathbf{K}) = n.$$

The same holds for $\phi(\mathbf{x})$, hence

$$\mathbb{E}_g[\log \phi(\mathbf{X})] = \mathbb{E}_\phi[\log \phi(\mathbf{X})].$$

$$\Rightarrow \int g(\mathbf{x}) \log \phi(\mathbf{x}) d\mathbf{x} = \int \phi(\mathbf{x}) \log \phi(\mathbf{x}) d\mathbf{x}.$$

Minimum Estimation Error

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Theorem (Estimation error)

For any random variable X and estimator $\hat{X}$, we have $\mathbb{E}(X - \hat{X})^2 \geq \frac{1}{2\pi e} e^{2h(X)}$ with equality iff X is Gaussian with mean $\hat{X}$.

Proof: $\mathbb{E}(X - \hat{X})^2 \geq \min_{\hat{X}} \mathbb{E}(X - \hat{X})^2 = \mathbb{E}(X - \mathbb{E}(X))^2 = \text{Var}(X) \geq \frac{1}{2\pi e} e^{2h(X)}.$

Corollary (Estimation error with side information)

Given side information Y and estimator $\hat{X}(Y)$, it follows that

$$\mathbb{E}(X - \hat{X}(Y))^2 \geq \frac{1}{2\pi e} e^{2h(X|Y)}.$$